

This assignment will not be graded and is only for practice.

Directional Derivatives:

Level 1:

1) Which of the following statements is(are) true?

1 point

- ☐ The directional derivative of $f(x, y)$ at a point (a, b) in the direction of a unit vector u is a non negative real number.
- ☐ The directional derivative of any function is always a scalar.
- ☐ The rate of change of $f(x_1, x_2, \dots, x_n)$ in the direction of a unit vector $u = (a_1, a_2, \dots, a_n)$ is called the directional derivative of f in the direction of u .
- ☐ The directional derivative of any function is a vector because it is dependent on the direction of a unit vector.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The directional derivative of any function is always a scalar.

The rate of change of $f(x_1, x_2, \dots, x_n)$ in the direction of a unit vector $u = (a_1, a_2, \dots, a_n)$ is called the directional derivative of f in the direction of u .

2) If the directional derivative of $f(x, y) = x + xy - 2y^2$ at the point $(2, 3)$ in the direction of the vector $(2\sqrt{2}, 2\sqrt{2})$ is $a\sqrt{2}$ where $a \in \mathbb{R}$, then find the value of a .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -3

1 point

3) If the directional derivative of $f(x, y, z) = xe^y + ye^z + ze^x$ at point $(1, 0, 1)$ in the direction of a vector $(2, -2, 1)$ is $k.e$ where $k \in \mathbb{R}$, then find the value of $6k$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

4) Suppose f is a multivariable function defined on domain $\mathbb{R}^2$. Which of the following is equal to the rate of change of f at point (x_1, y_1) in the direction of a vector $u = (u_1, u_2)$? **1 point**

- ☐ $\lim_{h \rightarrow 0} \frac{f((x_1, y_1) + h(u_1, u_2)) + f(x_1, y_1)}{h}$
- ☐ $\lim_{h \rightarrow 0} \frac{f((x_1, y_1) + h(u_1, u_2)) - f(x_1, y_1)}{h}$
- ☐ $\lim_{h \rightarrow 0} \frac{f((x_1, y_1) + h(\frac{u_1}{\sqrt{u_1^2 + u_2^2}}, \frac{u_2}{\sqrt{u_1^2 + u_2^2}})) - f(x_1, y_1)}{h}$
- ☐ $\lim_{h \rightarrow 0} \frac{f((x_1, y_1) + h(\frac{u_1}{\sqrt{u_1^2 + u_2^2}}, \frac{u_2}{\sqrt{u_1^2 + u_2^2}})) + f(x_1, y_1)}{h}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\lim_{h \rightarrow 0} \frac{f((x_1, y_1) + h(\frac{u_1}{\sqrt{u_1^2 + u_2^2}}, \frac{u_2}{\sqrt{u_1^2 + u_2^2}})) - f(x_1, y_1)}{h}$$

5) If the directional derivative of $f(x, y) = x \sin(y)$ at the point $(1, 0)$ in the direction of a unit vector (u_1, u_2) ($u_1, u_2 \in \mathbb{R}$) is 0.8, then find the absolute value of u_1 . (Enter your answer upto one decimal place only)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Range) 0.5,0.7

1 point

Level 2:

- 6) Consider the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as: $f(x, y) = \begin{cases} \frac{x^2 y}{x^4 + y^2} & x, y \neq 0 \\ 0 & x = y = 0 \end{cases}$ Find the directional derivative of f at $(0,0)$ in the direction of the vector $u = (\frac{\sqrt{3}}{2}, \frac{1}{2})$. (Enter your answer upto one decimal place only)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Range) 1.4,1.6

1 point

- 7) Suppose d is the directional derivative of a scalar valued multivariable function $f(x, y) = x^2 + y^2$ at a point $P(1, 2)$ in the direction of the line PQ , where point Q is at $(4, 6)$. The value of d is (Enter your answer upto one decimal place only)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Range) 4.3,4.5

1 point

- 8) Suppose $f(x, y, z) = ax^2y + yz^2 - z^2x$, where $a \in \mathbb{R}$, is a scalar valued multivariable function defined on domain $D \subset \mathbb{R}^3$. For what value of a , does the directional derivative of the given function at the point $(1, 1, 1)$ in the direction of the vector $u = (1, 2, -2)$ equal 3?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

- 9) The temperature at a point (x, y, z) is given by

$$T(x, y, z) = 10e^{-x^2+y^2-4z^2}$$

where T is measured in $^{\circ}C$ and x, y, z are measured in meters. If the rate of change of temperature at $(2, -1, 0)$ in the direction of a vector $u = (1, -1, 1)$ is $k \cdot \frac{e^{-3}}{\sqrt{3}}$ where $k \in \mathbb{R}$,

then the value of k is (approximate to two decimal places)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) -20

1 point

10) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be given by $f(x, y) = x^2 + y^2 - 2xy$. Suppose $D_u f(x_0, y_0)$ is the directional derivative at (x_0, y_0) in the direction of the unit vector $u = (\frac{1}{\sqrt{2}}, \frac{-1}{\sqrt{2}})$ and $D_v f(x_0, y_0)$ is the directional derivative at (x_0, y_0) in the direction of the unit vector $v = (\frac{1}{2}, \frac{\sqrt{3}}{2})$. Consider a matrix $A = \begin{bmatrix} D_u f(1, 2) & D_u f(2, 1) \\ D_v f(1, 2) & D_v f(2, 1) \end{bmatrix}$. What is the value of $\det(A)$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Your score is: 0/10